

Question 1: Harris Corner Detection for Satellite Image Registration

1.1 Limitations of Edge Detection in This Application

Edge detection algorithms (e.g., Sobel, Canny, Prewitt) identify pixels where intensity changes sharply, marking the boundaries of objects or regions. While useful for outlining shapes, edges are a poor basis for matching two independently captured satellite images for several reasons:

- **Lack of distinctiveness (aperture problem):** An edge point looks the same as its neighbours along the edge direction. A short segment of a straight edge (e.g., a road or field boundary) cannot be uniquely matched between two images because many points along that edge appear identical — only the position perpendicular to the edge is well constrained, not the position along it.
- **Sensitivity to illumination changes:** Satellite images of the same region taken at different times differ in sun angle, shadow length, and atmospheric haze. Edge strength and even edge presence change with lighting, so edges detected in one image often do not correspond to edges in the other, causing missed or spurious matches.
- **Sensitivity to orientation and viewpoint:** If the two images are rotated or slightly skewed relative to each other, straight-line edges shift direction, and simple edge-based correlation techniques fail to associate them correctly.
- **High density and redundancy:** Natural and urban scenes produce dense, continuous edge maps (vegetation boundaries, roads, coastlines). This redundancy makes it computationally expensive and ambiguous to search for correct correspondences, increasing the false-match rate.
- **No inherent localization in two dimensions:** Edges only constrain position in one direction (normal to the edge), so they cannot serve as precise, repeatable anchor points needed for accurate geometric alignment (registration) of images.
- **Background clutter and noise:** Differences in seasonal vegetation, cloud cover, and sensor noise introduce spurious edges that do not exist in the other image, further degrading matching reliability.

1.2 How Harris Corner Detection Improves the System

Harris Corner Detection identifies points where the intensity gradient changes significantly in more than one direction (i.e., corners or interest points), rather than along a single edge direction. It does this by analysing the local structure tensor (second-moment matrix) of image gradients within a small window and computing a corner response function R based on the eigenvalues of that matrix.

Improved Feature Localization

- **Two-dimensional constraint:** Because a corner has strong gradients in two directions, its position is well defined in both the x and y directions, unlike an edge point which is only constrained in one direction. This gives precise, repeatable pixel locations for registration.
- **High repeatability:** The same physical corner (e.g., a building corner, road intersection, field boundary junction) is very likely to be detected at the same location in both images, even when captured at different times, because corner structure is a stable geometric property of the scene rather than a lighting-dependent property.

Improved Matching Accuracy

- **Distinctive local structure:** Corners have unique local neighbourhoods (two dominant gradient directions), making them far easier to distinguish from one another than edge points, which reduces ambiguous or false correspondences during matching.
- **Fewer, higher-quality candidate points:** Harris detection typically yields a sparse set of highly distinctive points instead of a dense edge map, reducing the search space for correspondence matching and lowering the rate of mismatches.
- **Better geometric transformation estimation:** Accurate corner correspondences allow robust estimation of the transformation (rotation, translation, scale) needed to align the two images, such as through techniques like RANSAC combined with corner matches.

Improved Robustness Under Varying Imaging Conditions

- **Illumination invariance:** The Harris response is based on gradient direction changes rather than absolute intensity values, making it relatively robust to changes in brightness and contrast caused by different sun angles or atmospheric conditions between acquisition dates.
- **Rotation invariance:** The eigenvalue-based corner response is derived from the auto-correlation matrix, which is rotationally invariant — a corner remains detectable as a corner regardless of image orientation, unlike simple edge-orientation-based matching.
- **Resistance to background clutter:** Because Harris detection only responds strongly to true two-directional intensity variation, it suppresses flat regions and single-direction edges (e.g., uniform fields, cloud edges), focusing on stable, information-rich landmarks such as building corners and road intersections.
- **Noise suppression:** Use of a Gaussian smoothing window when computing the structure tensor reduces sensitivity to pixel-level sensor noise, improving reliability across images captured under different atmospheric or sensor conditions.

1.3 Summary

By shifting from edge-based matching to Harris Corner Detection, the remote sensing system replaces ambiguous, illumination-sensitive edge correspondences with a sparse set of distinctive, well-localized, and repeatable interest points. This directly improves the accuracy of feature correspondence, and — combined with descriptor matching and outlier rejection (e.g., RANSAC) — enables robust, precise image registration despite differences in illumination, orientation, and background content between images captured at different times.

Question 2: Image Representation for Real-Time Object Detection

2.1 Overview of the Problem

An autonomous surveillance system must detect vehicles and pedestrians in real time from high-resolution color video. Processing full-color (RGB) images directly increases computational load because each pixel carries three channels of data, and most classical feature-extraction and edge-detection algorithms must either process each channel separately or first derive intensity information from them. To meet real-time constraints, the engineers must choose the image representation (color, grayscale, or binary) that best balances computational efficiency with detection accuracy and reliability.

2.2 Comparative Analysis of Image Representations

Representation	Computational Efficiency	Edge Detection Performance	Feature Extraction Suitability
Color (RGB)	Lowest efficiency — 3 channels to process, ~3x memory & computation of grayscale	Requires per-channel gradients or conversion; richer but slower	Best discriminative power (uses hue/saturation), but costly for real-time use
Grayscale	High efficiency — single intensity channel, ~3x faster than RGB	Well suited; most edge operators (Sobel, Canny) designed for single-channel intensity	Good balance — preserves shape, texture, gradient information needed for detection
Binary	Highest efficiency — 1 bit/pixel, fastest processing and lowest memory	Poor — thresholding destroys gradient/texture detail; edges become jagged or lost	Weak — loses shape detail and is highly sensitive to lighting/threshold selection

2.3 Impact on Computational Efficiency

- **Color images:** Carry three times the data of grayscale images (R, G, B channels), increasing memory bandwidth, storage, and processing time for every subsequent operation (filtering, convolution, feature matching), which directly conflicts with real-time frame-rate requirements.
- **Grayscale images:** Reduce data volume to a single intensity channel, cutting computation roughly threefold for most pixel-wise and convolutional operations, enabling faster frame processing suitable for real-time video.
- **Binary images:** Offer the smallest data footprint (a single bit per pixel) and the fastest processing, but this efficiency comes from throwing away almost all intensity and texture information, which severely limits usefulness for complex detection tasks.

2.4 Impact on Edge Detection

- **Color images:** Edge detectors are typically defined for scalar (single-channel) images, so applying them to color requires either converting to grayscale first or computing gradients per channel and combining them — the latter adding further computational overhead for often marginal gains in edge quality.

- **Grayscale images:** Standard gradient-based operators (Sobel, Prewitt, Canny) operate directly and efficiently on grayscale intensity, producing reliable, well-defined edges that correspond closely to object boundaries such as vehicle outlines and pedestrian silhouettes.
- **Binary images:** Since binary conversion applies a hard threshold, subtle intensity gradients are lost. Edges become coarse, jagged, and highly sensitive to the chosen threshold and to lighting conditions, often merging or fragmenting object boundaries incorrectly.

2.5 Impact on Feature Extraction Performance

- **Color images:** Enable color-based cues (e.g., distinguishing a red vehicle from background foliage) that can aid detection but are more susceptible to illumination and shadow variation, and require heavier descriptors (e.g., color histograms) that slow down real-time pipelines.
- **Grayscale images:** Preserve the structural, textural, and gradient information that most classical and even many deep-learning feature extractors (HOG, SIFT, edge/corner detectors, CNN feature maps) rely on, while significantly reducing computational cost compared to color.
- **Binary images:** Suitable mainly for simple silhouette-based analysis (e.g., blob detection in controlled, high-contrast scenes) but perform poorly under varying illumination, shadows, or cluttered backgrounds typical of outdoor surveillance, since fine shape and texture cues needed to distinguish vehicles from pedestrians are lost.

2.6 Justification of the Most Appropriate Representation

For an autonomous surveillance system requiring real-time detection of vehicles and pedestrians under varying outdoor conditions, grayscale representation is the most appropriate choice, for the following reasons:

- It offers a substantial reduction in computational load (roughly threefold over color) sufficient to meet real-time processing requirements, unlike full RGB processing.
- It retains enough structural detail (edges, gradients, texture, and shape) — unlike binary images — to reliably distinguish object classes such as vehicles and pedestrians and to support standard feature descriptors (HOG, SIFT/SURF, corner/edge detectors).
- It is compatible with virtually all classical edge-detection and feature-extraction algorithms, which are designed for single-channel intensity input, avoiding the overhead of multi-channel or per-channel processing.
- It is comparatively robust to color-based illumination artifacts (e.g., color casts from streetlights) that can confuse color-based detectors, while binary thresholding would be far more fragile under such variable lighting.

Where color information provides a meaningful discriminative advantage (e.g., detecting brake lights or specific uniform colors) and computational budget allows, a hybrid approach can be used: primary detection and tracking performed on grayscale (or a lightweight luminance-derived representation) for speed, with selective color analysis applied only to regions of interest already flagged by the grayscale pipeline. This preserves real-time performance while retaining the option to exploit color cues when needed.

2.7 Summary

Color images offer the richest information but are computationally expensive; binary images are fastest but discard too much detail for reliable detection under real-world conditions. Grayscale representation provides the best trade-off between computational efficiency and feature/edge information richness, making it the most suitable representation for accurate, reliable, real-time vehicle and pedestrian detection in autonomous surveillance systems.